

On the Existence of a Solution to the Coupled HJB - KFP System

Avi Wine

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1 Introduction

(Italics are copied) Mean field games (MFG), which were introduced by Lasry and Lions [24, 25, 26] and independently by Huang, Caines, and Malham'e [22], model Nash equilibria of rational games of stochastic optimal control in which there are infinitely many players. The equilibria are characterized by a system of partial differential equations (PDE) that consist of a Hamilton–Jacobi–Bellman (HJB) equation coupled with a Kolmogorov–Fokker–Planck (KFP) equation. Comprehensive reviews of MFG, including analysis and applications, are well-documented in the literature [3, 19, 20]. We consider as a model problem a quasilinear elliptic stationary MFG system of the form

$$\Delta u + H(x, \nabla u) + \lambda u = f(x, m) \text{ in } \Omega,$$

$$\Delta m + m \frac{\partial H}{\partial p}(x, \nabla u) + \lambda m = G(x) \text{ in } \Omega,$$

along with homogeneous Dirichlet boundary condition $u = 0$ and $m = 0$ on $\partial\Omega$. These equations can be interpreted as ... with $m(x)$ representing player density and $u(x)$ the *value function of the underlying optimal control problem faced by the players*. Here $\frac{\partial H}{\partial p}$ is shorthand for the partial derivative w.r.t ∇u .

In this paper I demonstrate the existence of a solution pair (u, m) in $H_0^1(\Omega)$, provided the following conditions are met:

1. Ω is ...
2. $H(x, \nabla u) : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ is

- Lipschitz continuous with respect to ∇u ,
 - bounded by $C(|x| + |\nabla u|)$ for a given constant C and for all $x \in \mathbb{R}, \nabla u \in \mathbb{R}^n$.
3. The partial derivative H_p is continuous with respect to ∇u and essentially bounded by some constant L .
 4. λ is sufficiently large.
 5. $f(x, m(x)) : (\Omega, L^2(\Omega)) \rightarrow L^2(\Omega)$ and is L^2 -continuous.
 6. $G(x) \in L^2(\Omega)$.

2 Method

Our/My? solution consists of two stages. First we'll demonstrate solvability for each equation -while fixing the second free variable-independent of the other. Next comes the existence of a solution pair that solves both equations simultaneously.

2.1 Independant Solutions

2.1.1 Solving Kolmogorov–Fokker–Planck

To solve $\Delta m + m \frac{\partial H}{\partial p}(x, \nabla u) + \lambda m = G(x)$ with fixed $u(x)$, we call upon the famous Lax–Milgram Thm.

Theorem 1 (Lax–Milgram). *Let V be a Hilbert space, let $a : V \times V \rightarrow \mathbb{R}$ be a*

1. *bounded:* $|a(u, v)| \leq c_1 \|u\| \|v\|$
2. *coercive:* $a(v, v) \geq c_0 \|v\| \|v\|$

bilinear form for all $u, v \in V$, and let $l : V \rightarrow \mathbb{R}$ be a bounded linear functional. Then there exists a unique $u \in V$ such that

$$a(u, v) = l(v) \text{ for all } v \in V$$

Furthermore the solution $\|u\| \leq \frac{1}{c_0} \|l(v)\|$

Note that this will only work for (KFP) which becomes a linear PDE in m once $u \in H_0^1(\Omega)$ is fixed.

Our (KFP) bilinear form is It's bounded... condition (3)... and coercive ... condition (4). $l(v)$ functional is just L1 norm of G , which is bounded ... condition (6)

For future use, let

$$S : H_0^1(\Omega) \rightarrow H_0^1(\Omega)$$

$$S(u) = m \text{ such that } m \text{ solves (KFP) for given } u$$

We know S is a well defined map since said m both exists and is unique.

2.1.2 Solving Hamilton–Jacobi–Bellman

Lax-Milgram will unfortunately not do to solve $\Delta u + H(x, \nabla u) + \lambda u = f(x, m)$ in Ω , for fixed $m(x)$ since it doesn't reduce to a linear PDE. We must therefore resolve to adopt the following fixed point theorem.